Mathematical Foundations of Political Methodology

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Day 3: Random Variables: Discrete

- Random Variables
- Expectation, Variance
- Discrete Probability Mass Function
- Cumulative Distribution Functions
- Joint Probability Mass Functions

Random Variables

Given a sample space Ω, and a probability law P:

Definition

A random variable is a function that assigns real numbers (usually) to events in a sample space Ω*.*

$$X(\omega) : \Omega \to \mathbb{R}$$

Random Variables: Example

Sample space from tossing three coins:

$$\Omega = \{HHH,\ HHT,\ HTH,\ THH,\ HTT,\ THT,\ TTH,\ TTT\}$$

Let $Y(\omega)$ be the number of heads in outcome $\omega \in \Omega$.

- $Y(HHH) = 3$
- $Y(HHT) = Y(HTH) = Y(THH) = 2$
- $Y(HTT) = Y(THT) = Y(TTH) = 1$
- $Y(TTT) = 0$

Random Variables: Example II

Sample space from the number of dots on a die roll.

$\Omega = \{$⚀, ⚁, ⚂, ⚃, ⚄, ⚅$\}$

Let $Z(\omega)$ be the dots on the die face $\omega \in \Omega$.

- Z(⚀)=1
- Z(⚁)=2
- Z(⚂)= 3
- Z(⚃)=4
- Z(⚄)=5
- Z(⚅)=6

Random Variables: Example III

Sample space from the number of dots on a die roll.

$\Omega = \{$, , , , , $\}$

Let $X(\omega)$ be 3 to the power of the number of dots on the die face $\omega \in \Omega$ minus 1.

- X()$= 3^{1-1} = 3^0 = 1$
- X()$= 3^{2-1} = 3$
- X()$= 3^{3-1} = 9$
- X()$= 3^{4-1} = 27$
- X()$= 3^{5-1} = 81$
- X()$= 3^{6-1} = 243$

$X(\omega) = 3^{\omega-1}$

functions

- Random Variables are functions.
- In R, we see functions look like a word followed by ():
  - mean(), median(), head(), str()
- Our own functions are written using:

```
function(input){
    do something
    return(output)}
```

Properties of Random Variables

Random variables (for us) will always give a number.

- X,Y, Z: capital letters for a random variable.
- x, y, z: are the output of the function.

Kinds of Random Variables

Definition

A random variable may be

1. *discrete: if outcomes are countable.*
2. *continuous: If outcomes can be any value in some interval.*
3. *mixed: if outcomes are a combination of events and intervals.*

Connecting random variables to probability

- X assigns some numbers to events.
- Remember, probability assigns a likelihood to an event.
- What is the probability that $X(\omega)$ gives some number x?
- $P(X = x) \equiv P(\{\omega \in \Omega : X(\omega) = x\})$
- This is another function (of a function)

Discrete Probability Function

Definition

If X is a discrete random variable, the probability of each outcome x is provided by a probability function:

$$p_X(x) = P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\})$$

- this is also called the *probability mass function* (pmf)
- We often just write $p(x)$, some people use $f_X(x)$

Building a Discrete Probability Function: Example

- Sample space: $\Omega = \{\text{Pashto, Not Pashto}\}$
- X=1 if ω = Pashto
- X=0 if ω = Not Pashto
- P(X=1)=.3

A pmf can take on any value of x, so we must consider all possibilities of x.

- $x = 1$
- $x = 0$
- all other values of x.

- If $x = 1$

$$P(X = x) = P(X = 1) = P(\text{Pashto}) = p = .3$$

- If $x = 0$

$$P(X = x) = P(X = 0) = P(\text{Not Pashto}) = 1 - p = .7$$

- all other values of x.

$$P(X = x) = 0$$

Combining these results, we get the probability function.

$$f_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

This can be written shorthand as:

$$f_X(x) = p^x(1 - p)^{(1-x)}, \quad x \in \{0, 1\}$$

Where the 0 is left implicit.

Dichotomous process

To summarize

- If X only has two values 0 and 1 and $p = P(X = 1)$ then X has the probability function:

$$f_X(x) = p^x(1-p)^{(1-x)}, \quad x \in \{0, 1\}$$

- This is called a Bernoulli distribution:

$$X \sim \text{Bernoulli}(p)$$

- p is a *parameter*

Bernoulli Random Variable

Suppose we flip a fair coin and $Y = 1$ if the outcome is Heads .

$$\begin{aligned}
Y &\sim \text{Bernoulli}(1/2) \\
p(1) &= (1/2)^1(1-1/2)^{1-1} = 1/2 \\
p(0) &= (1/2)^0(1-1/2)^{1-0} = (1-1/2)
\end{aligned}$$

Example: Winning a War

Suppose country 1 is engaged in a conflict and can either win or lose.
Define $Y = 1$ if the country wins and $Y = 0$ otherwise.
Then,

$$Y \sim \text{Bernoulli}(\pi)$$

Suppose country 1 is deciding whether to fight a war.
Engaging in the war will cost the country c.
If they win, country 1 receives B.
What is 1's expected utility from fighting a war?

$$\begin{aligned}
E[U(\text{war})] &= (\text{Utility}|\text{win}) \times P(\text{win}) + (\text{Utility}|\text{lose}) \times P(\text{lose}) \\
&= (B - c)P(Y = 1) + (-c)P(Y = 0) \\
&= B \times p(Y = 1) - c(P(Y = 1) + P(Y = 0)) \\
&= B \times \pi - c
\end{aligned}$$

Example: Sequence of Bernoulli trials

We might be interested in the outcome of several independent trials **ie. with replacement**.

Consider a coin, flipped twice. Suppose $X_1(T) = 0$, $X_1(H) = 1$, $X_2(T) = 0$, $X_2(H) = 1$

$$P(X_1 = 1,\ X_2 = 1) = P(X_1 = 1 \cap X_2 = 1) = P(X_1 = 1)P(X_2 = 1)$$

$$p(x_1, x_2) = p^{x_1}(1-p)^{1-x_1}p^{x_2}(1-p)^{1-x_2}$$

$$p(x_1, x_2) = p^{x_1+x_2}(1-p)^{2-x_1-x_2}$$

Let $X_1, ..., X_n$ be independent Bernoulli random variables with the same parameter p.

$$p(x_1, ...x_n) = p(x_1)p(x_2)...p(x_n) = p^{x_1+...x_n}(1-p)^{n-x_1-....-x_n}$$

$$\text{for } x_i \in \{0, 1\} \text{ and } i = 1, ..., n$$

Expectation

What can we expect from a trial?
Value of random variable for any outcome weighted by the probability of observing that outcome
probability weighted average

Definition

Expected Value: define the expected value of a function X as,

$$E[X] = \sum_{x:p(x)>0} xp(x)$$

In words: for all values of x with $p(x)$ greater than zero, take the weighted average of the values

Expectation informally

The expectation is based on the principle that the value of a future gain should be directly proportional to the chance of getting it.

$$E[X+Y] = E[X] + E[Y]$$
$$E[a] = a$$
$$E[aX] = aE[X]$$
$$E[E[X]] = E[X]$$

$$E[XY] \neq E[X] \times E[Y]$$

Properties of the Expecation

$$E[a] = a$$

Proof: Suppose Y is a random variable = a with probability 1 and 0 otherwise:

$$E[Y] = \sum_{y:p(y)>0} yp(y) = a * 1$$

Also:

$$E[aX] = \sum_{x:p(x)>0} a \ x \ p(x)$$

$$E[aX] = a \sum_{x:p(x)>0} x \ p(x)$$

Indicator Variables and Probabilities

Proposition

Suppose A is an event. Define random variable I such that $I = 1$ if an outcome in A occurs and $I = 0$ if an outcome in A^c occurs. Then,

$$E[I] = P(A)$$

Proof.

$$\begin{aligned} E[I] &= 1 \times P(A) + 0 \times P(A^c) \\ &= P(A) \end{aligned}$$

$\square$

Variance

Expected value is a measure of central tendency.
What about spread? Variance

- For each value, we might measure distance from center
  - Euclidean distance, squared $d(x, E[x])^2 = (x - E[x])^2$
- Then, we might take weighted average of these distances,

$$
\begin{aligned}
E[(X - E[X])^2] &= \sum_x (x - E[X])^2 p(x) \\
&= \sum_x \left(x^2 - E[X]x - xE[X] + E[X]^2\right) p(x) \\
&= \sum_x x^2 p(x) - \sum_x 2xE[X]p(x) + \sum_x E[X]^2 p(x) \\
&= \sum_x x^2 p(x) - 2E[X] \sum_x xp(x) + \sum_x E[X]^2 p(x) \\
&= E[X^2] - 2E[X]^2 + E[X]^2 \\
&= E[X^2] - E[X]^2 \\
&= \text{Var}(X)
\end{aligned}
$$

Variance

Definition

The variance of a random variable X, $var(X)$, is

$$\begin{aligned} var(X) &= E[(X - E[X])^2] \\ &= E[X^2] - E[X]^2 \end{aligned}$$

- We will define the standard deviation of X, $\text{sd}(X) = \sqrt{\text{var}(X)}$
- $\text{var}(X) \geq 0$.

Variance Corollary

Corollary

$Var(aX + b) = a^2 Var(X)$

Proof.

Define $Y = aX + b$. Now, we know that
$Var(Y) = E[(Y - E[Y])^2]$. Let's substitute and use our other corollary

$$\begin{aligned} Var(Y) &= E[(aX + b - aE[X] - b)^2] \\ &= E[(a^2X^2 - 2a^2XE[X] + a^2E[X]^2)] \\ &= a^2E[X^2] - 2a^2E[X]^2 + a^2E[X]^2 \\ &= a^2(E[X^2] - E[X]^2) \\ &= a^2Var(X) \end{aligned}$$

□

Bernoulli Random Variable Moments

Suppose $Y \sim \text{Bernoulli}(\pi)$

$$
\begin{aligned}
E[Y] &= 1 \times P(Y = 1) + 0 \times P(Y = 0) \\
&= \pi + 0(1 - \pi) = \pi \\
\text{var}(Y) &= E[Y^2] - E[Y]^2 \\
E[Y^2] &= 1^2 P(Y = 1) + 0^2 P(Y = 0) \\
&= \pi \\
\text{var}(Y) &= \pi - \pi^2 \\
&= \pi(1 - \pi)
\end{aligned}
$$

$E[Y] = \pi$
$\text{var}(Y) = \pi(1 - \pi)$ What is the maximum variance?

Trials with More than Two Outcomes

Definition

Suppose we observe a trial, which might result in J outcomes.
And that $P(outcome\ = i) = \pi_i$
$\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_J)$ *where* $Y_j = 1$ *if outcome j occurred and 0 otherwise.*
Then $\boldsymbol{Y}$ *follows a* *multinomial* *distribution, with*

$$p(\boldsymbol{y}) \quad = \quad \pi_1^{y_1}\pi_2^{y_2}\ldots\pi_k^{y_k}$$

if $\sum_{i=1}^{k} y_i = 1$ *and the pmf is* 0 *otherwise.*
Equivalently, we'll write

$$\begin{aligned} \boldsymbol{Y} &\sim \textit{Multinomial}(1, \boldsymbol{\pi}) \\ \boldsymbol{Y} &\sim \textit{Categorial}(\boldsymbol{\pi}) \end{aligned}$$

Multinomial Properties + Notes

Computer scientists: commonly call Multinomial$(1, \boldsymbol{\pi})$ Discrete$(\boldsymbol{\pi})$.

$$\begin{aligned} E[X_i] &= N\pi_i \\ \text{var}(X_i) &= N\pi_i(1 - \pi_i) \end{aligned}$$

Counting the Number of Events

Often interested in counting number of events that occur:

1) Number of wars started
2) Number of speeches made
3) Number of bribes offered
4) Number of people waiting for license

Generally referred to as event counts

Stochastic processes: a course provide introduction to many processes (Queing Theory)

Definition

Suppose X is a random variable that counts the number of successes in N independent and identically distributed Bernoulli trials. Then X is a Binomial random variable,

$$p(k) \;=\; \binom{N}{k}\pi^k(1-\pi)^{1-k}$$

for $k \in \{0, 1, 2, \ldots, N\}$ and $p(k) = 0$ otherwise.
Equivalently,

$$Y \;\sim\; Binomial(N, \pi)$$

Binomial Random Variable Moments

$Z = \sum_{i=1}^{N} Y_i$ where $Y_i \sim \text{Bernoulli}(\pi)$

$$
\begin{aligned}
E[Z] &= E[Y_1 + Y_2 + Y_3 + \ldots + Y_N] \\
&= \sum_{i=1}^{N} E[Y_i] \\
&= N\pi \\
\text{var}(Z) &= \sum_{i=1}^{N} \text{var}(Y_i) \\
&= N\pi(1-\pi)
\end{aligned}
$$

$E[Z] = N\pi$
$\text{var}(Z) = N\pi(1-\pi)$

Example: Counts

We might be interested in the *sum* of several independent trials:

- n draws from a population size N with replacement.
- If Y is number of the sample that have a binary characteristic (support Trump, show up heads, etc).
- and M is the number of the people in the population that have a binary characteristic.

Our random variable, Y, is equal to the sum of successes. Let y be an integer between 0 and n,

$$P(Y = y) = \binom{n}{y} p^y (1 - p)^{(n-y)}$$

where $p = \frac{M}{N}$

Binomial Random Variable

- A model to count the number of successes across N trials
  - Assume the Bernoulli trials are independent
  - Each Bernoulli trial i is

$$Y_i \sim \text{Bernoulli}(\pi)$$

    Independent and identically distributed.
- $Z =$ number of successful trials
- Derive probability mass function $P(Z = M) = p(M)$
- One way to obtain M successful trials:

$$
\begin{aligned}
P(Y_1 = 1, Y_2 = 0, Y_3 = 1, \ldots, Y_N = 1) \\
&= P(Y_1 = 1)P(Y_2 = 0)\cdots P(Y_N = 1) \\
&= \underbrace{P(Y_1 = 1)P(Y_3 = 1)\cdots P(Y_z = 1)}_{M} \times \underbrace{P(Y_2 = 0)\cdots P(Y_N = 0)}_{N-M} \\
&= \underbrace{\pi\pi\cdots\pi}_{M} \times \underbrace{(1-\pi)(1-\pi)\cdots(1-\pi)}_{N-M} \\
&= \pi^M (1-\pi)^{N-M}
\end{aligned}
$$

Are we done? No

- This is just one instance of M successes
- How many total instances?
  - N total trials
  - We want to select M
- $\binom{N}{M} = \frac{N!}{(N-M)!M!}$

Then,

$$P(Z = M) \;=\; p(M) = \binom{N}{M}\pi^M(1-\pi)^{N-M}$$

Origins of Binomial

One head on one Coin:

$$P(Y = 1) = p$$

One head on two coins:

$$P(Y = 1) = p(1 - p) + (1 - p)p = 2 * p(1 - p)^1$$

One head on three coins:

$$P(Y = 1) = p(1-p)(1-p)+(1-p)p(1-p)+(1-p)(1-p)p = 3*p(1-p)^2$$

One head on four coins:

$$P(Y = 1) = p(1 - p)^3 + (1 - p)p(1 - p)^2$$

$$+(1 - p)^2 p(1 - p) + (1 - p)^3 p = 4 * p(1 - p)^3$$

Origins of Binomial

One head on four coins

$$P(Y = 1) = 4 * p(1 - p)^3$$

Two heads on four coins

$$P(Y = 2) = 6 * p^2(1 - p)^2$$

Three heads on four coins

$$P(Y = 3) = 4 * p^3(1 - p)$$

Number of different realizations with y successes: $\binom{n}{y}$

$$P(Y = y) = \binom{n}{y} p^y (1 - p)^{(n-y)}$$

where $p = \frac{M}{N}$

Origins of Binomial

If $X_1, ...X_n$ are Bernoulli distributed with probability p, then

$$Y = \sum_{k=1}^{n} X_k \sim \text{Binomial}(n, p)$$

n and p are parameters of the binomial.

Assumptions of Binomial Distribution

- The number of trials is fixed,
- The probability of success is the same for each trial.
- The trials are independent

Example: Baseball!

- The World Series of Baseball ends when one team wins 4 games.
- Each game is a Bernoulli trail.
- Length of World Series $\in \{4, 5, 6, 7\}$.
- What is the probability that the World Series lasts 7 games?
- P(X=7)?

Example: Counts in a time period

We might be interested in the sum of events occurring in a specific time or space.

- Number of wars in a year.
- Number of diseased villages in a district.

Let X be the number of times an event occurs in a unit of time or space.

- The probability that the event occurs in a given unit of time or space is the same for all the units.
- The number of events that occur in one unit of time is independent of the number of events in other units.

Then X is a Poisson random variable, with rate $= \lambda$ and pf:

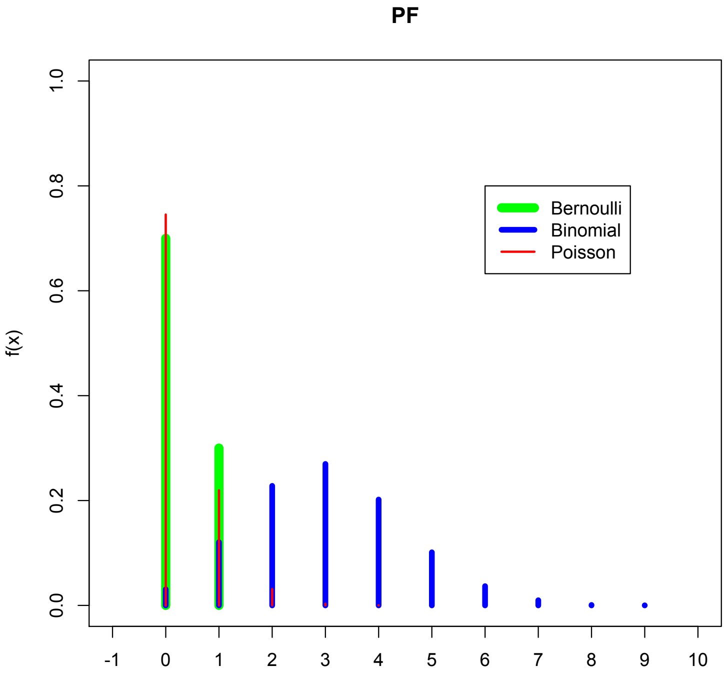
$$p(x) = \frac{\lambda^x}{x!} e^{-\lambda} \quad x = 0, 1, 2...$$

Example: Poisson Distribution

If, on average, two customers sign up every hour, what is the probability that 4 customers show up in a given hour?

$$\lambda = 2$$

$$P(X = 4) = \frac{2^4}{4!} e^{-2} = .09$$


PF
f(x)
0.0
0.2
0.4
0.6
0.8
1.0
-1
0
1
2
3
4
5
6
7
8
9
10
Bernoulli
Binomial
Poisson

Functions of Random Variables

Definition

If $Y = g(X)$, and X is a random variable with probability mass function $p_X(x)$, Y is a random variable with pmf:

$$p_Y(y) = \sum_{\{x|g(x)=y\}} p_X(x)$$